WORKED SOLUTION

STEP II 2023, Question 1

Pure • Integration by Substitution

STEP II 2023, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The substitution $x = 1/t$ swaps $\frac{1}{2}$ and 2, so it maps the interval $[\frac{1}{2}, 2]$ onto itself (backwards). An integral over this interval can therefore be rewritten as a *different-looking* integral over the *same* interval. Sometimes the new integrand is easy to integrate directly; sometimes it is a partner of the old one, so that adding the two (or comparing them) does the work for us.

Solution

Part (i)

Put $x = 1/t$, so $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$. Since $a > 0$, every x in $[a, b]$ is positive, and so is the corresponding t. Then

$$1 + x^2 = \frac{t^2 + 1}{t^2} \quad \Longrightarrow \quad (1 + x^2)^{3/2} = \frac{(1 + t^2)^{3/2}}{t^3},$$

where we used $\sqrt{t^2} = t$ because $t > 0$. The limits $x = a$ and $x = b$ become $t = a^{-1}$ and $t = b^{-1}$. Hence

$$\int_a^b \frac{1}{(1 + x^2)^{3/2}}\,\mathrm{d}x = \int_{a^{-1}}^{b^{-1}} \frac{t^3}{(1 + t^2)^{3/2}} \cdot \left(-\frac{1}{t^2}\right)\mathrm{d}t = \int_{a^{-1}}^{b^{-1}} \frac{-t}{(1 + t^2)^{3/2}}\,\mathrm{d}t,$$

as required.

Part (ii)

(a) With $a = \frac{1}{2}$ and $b = 2$, the new limits are 2 and $\frac{1}{2}$. Swapping them absorbs the minus sign:

$$\int_{1/2}^{2} \frac{1}{(1 + x^2)^{3/2}}\,\mathrm{d}x = \int_{1/2}^{2} \frac{t}{(1 + t^2)^{3/2}}\,\mathrm{d}t.$$

The new integrand is (a multiple of) the derivative of a power of $1+t^2$: since $\frac{\mathrm{d}}{\mathrm{d}t}(1+t^2)^{-1/2} = -t(1+t^2)^{-3/2}$,

$$\int_{1/2}^{2} \frac{t}{(1+t^2)^{3/2}}\,\mathrm{d}t = \Big[-(1+t^2)^{-1/2}\Big]_{1/2}^{2} = -\frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5/4}} = -\frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}} = \frac{1}{\sqrt{5}}.$$

(b) The integrand is even, so the integral over $[-2,2]$ is twice the integral over $[0,2]$. Split this as $[0,\frac{1}{2}] \cup [\frac{1}{2},2]$; the second piece is part (a).

? **Ask yourself** Part (i) needs $a > 0$. How can we handle the piece starting at 0?

Work on $[a,\frac{1}{2}]$ with $0 < a < \frac{1}{2}$, where part (i) does apply, and then let $a \to 0^+$. Part (i) with $b = \frac{1}{2}$ gives

$$\int_{a}^{1/2} \frac{1}{(1+x^2)^{3/2}}\,\mathrm{d}x = \int_{2}^{1/a} \frac{t}{(1+t^2)^{3/2}}\,\mathrm{d}t = \frac{1}{\sqrt{5}} - \frac{1}{\sqrt{1+a^{-2}}}.$$

As $a \to 0^+$, $a^{-2} \to \infty$ and the last term tends to 0. The left-hand side tends to the integral over $[0,\frac{1}{2}]$, since the integrand is continuous there. So that piece is $\frac{1}{\sqrt{5}}$ too, and

$$\int_{-2}^{2} \frac{1}{(1+x^2)^{3/2}}\,\mathrm{d}x = 2\Big(\frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5}}\Big) = \frac{4}{\sqrt{5}}.$$

Part (iii)

(a) Write $J = \displaystyle\int_{1/2}^{2} \frac{1}{(1+x^2)^2}\,\mathrm{d}x$ and apply $x = 1/t$ again. Now $(1+x^2)^2 = (1+t^2)^2/t^4$, so

$$J = \int_{2}^{1/2} \frac{t^4}{(1+t^2)^2} \cdot \Big(-\frac{1}{t^2}\Big)\mathrm{d}t = \int_{1/2}^{2} \frac{t^2}{(1+t^2)^2}\,\mathrm{d}t,$$

which, after renaming the dummy variable t as x, is the first equality. Call this second integral K, so $J = K$. Adding the two integrands,

$$J + K = \int_{1/2}^{2} \frac{1+x^2}{(1+x^2)^2}\,\mathrm{d}x = \int_{1/2}^{2} \frac{1}{1+x^2}\,\mathrm{d}x,$$

and since $J = K$ this gives $J = \frac{1}{2}\displaystyle\int_{1/2}^{2} \frac{1}{1+x^2}\,\mathrm{d}x$, completing the chain of equalities.

Hence

$$J = \tfrac{1}{2}(\arctan 2 - \arctan \tfrac{1}{2}).$$

This can be tidied in either of two ways. Since $\arctan 2 + \arctan \frac{1}{2} = \frac{\pi}{2}$ (complementary angles in a right-angled triangle with legs 1 and 2), $J = \frac{\pi}{4} - \arctan \frac{1}{2}$. Or, by the subtraction formula for tan, $\tan(\arctan 2 - \arctan \frac{1}{2}) = \dfrac{2 - \frac{1}{2}}{1+1} = \frac{3}{4}$, and the difference lies in $(0, \frac{\pi}{2})$, so

$$\int_{1/2}^{2} \frac{1}{(1+x^2)^2}\,\mathrm{d}x = \tfrac{1}{2}\arctan\tfrac{3}{4} \approx 0.3218.$$

(b) Let L be the required integral. Before reaching for a hard antiderivative, try $x = 1/t$ once more. The integrand becomes

$$\frac{1-\frac{1}{t}}{\frac{1}{t}\sqrt{\frac{1+t^2}{t^2}}} = \frac{\frac{t-1}{t}}{\frac{\sqrt{1+t^2}}{t^2}} = \frac{t(t-1)}{\sqrt{1+t^2}},$$

using $t > 0$ again. Multiplying by $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$ and noting that the limits run from 2 to $\frac{1}{2}$,

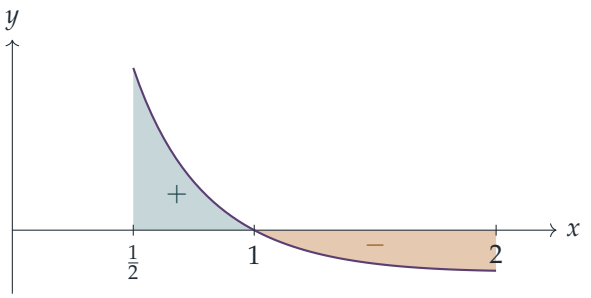
$$L = \int_2^{1/2} \frac{-(t-1)}{t\sqrt{1+t^2}}\,\mathrm{d}t = \int_{1/2}^{2} \frac{t-1}{t\sqrt{1+t^2}}\,\mathrm{d}t = -\int_{1/2}^{2} \frac{1-t}{t\sqrt{1+t^2}}\,\mathrm{d}t = -L.$$

? **Ask yourself** The substitution has turned L into $-L$. Is that a contradiction?

No: it just tells us the value. From $L = -L$ we get $2L = 0$, so

$$\int_{1/2}^{2} \frac{1-x}{x(1+x^2)^{1/2}}\,\mathrm{d}x = 0.$$

The picture agrees: the positive area over $[\frac{1}{2}, 1]$ exactly cancels the negative area over $[1, 2]$, because $x \mapsto 1/x$ swaps those two pieces and, by the working above, carries the area of one onto minus the area of the other.



Checks

- An antiderivative of $(1+x^2)^{-3/2}$ is $x(1+x^2)^{-1/2}$ (differentiate to confirm). It gives $\frac{2}{\sqrt{5}} - \frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \approx 0.4472$ for (ii)(a) and $\frac{4}{\sqrt{5}} \approx 1.7889$ for (ii)(b), matching our answers.
- In (iii)(a), $\frac{1}{2}\arctan\frac{3}{4} \approx 0.32175$ and $\frac{\pi}{4} - \arctan\frac{1}{2} \approx 0.32175$ agree. Numerical quadrature of all three integrals in the chain gives the same value.
- In (iii)(b) the integrand is positive for $x < 1$ and negative for $x > 1$, so a zero answer is plausible. Numerical quadrature gives 0 to within rounding error.

Common slips

- Losing the minus sign from $\mathrm{d}x = -t^{-2}\,\mathrm{d}t$, or forgetting that it is cancelled by swapping the limits.

- In (ii)(b), applying part (i) with $a = 0$, which would need the limit $t = 1/0$. Use a limit, or simply the antiderivative.
- Writing $\sqrt{(1+t^2)/t^2} = \sqrt{1+t^2}/t$ without noting that this needs $t > 0$.

Beyond STEP

The map $x \mapsto 1/x$ is a symmetry of the interval $[\frac{1}{2}, 2]$, and exploiting the symmetries of an integral is a recurring trick, as in $\int_0^{\pi/2} \ln \sin x \, \mathrm{d}x$ via $x \mapsto \frac{\pi}{2} - x$. Part (ii)(b) is a first taste of an improper integral: a substitution sent a finite endpoint to infinity, and we had to take a limit carefully. Abbott's *Understanding Analysis* treats limits of integrals rigorously, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with integration.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.